Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION – JUNE 2025

Name of the Course: **M.Tech** Semester: 2

Name of the Paper: **Digital System Design using Verilog** Paper Code: **VDM 202**

Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	**Analyze the following Verilog code and answer:** **1. What is the output waveform of clk and out?** **2. At which simulation times does out change?** **module clk_out (clk, out);** **output reg clk, out;** **initial begin** **clk = 1'b0;** **out = 1'b0;** **$monitor($time, " clk = %b, out = %b", clk, out);** **#50 $finish;** **end** **always #5 clk = ~clk;** **always @(posedge clk) begin** **if (out == 1'b0)** **out <= 1'b1;** **else** **out <= 1'b0;** **end** **endmodule**	CO1
(b)	**Explain the following data types in Verilog HDL:-** **1. Reg** **2. Wire** **3. Integer** **4. Real** **5. time**	
(c)	**Explain the key lexical conventions in Verilog HDL.**	
Q2	(20 marks)	
(a)	**Write the Verilog description of 4-bit Subtractor using** **i. Gate level modelling** **ii. Data flow modelling**	CO2, CO3

	iii. Behavioral modelling	
(b)	Analyze the following Verilog code and answer: module operator_test; reg [3:0] x = 4'b1011, y = 4'b1101; reg [7:0] z; initial begin #10 z = {x, y}; #10 z = x << 2; #10 z = (x > y) ? x : y; #10 z = x & ~y; #10 z = |y; #10 $finish; end initial begin $monitor("Time = %t: z = %b", $time, z); end endmodule What is the output value of z at each time step?	
(c)	With the help of suitable examples, explain the differences between FUNCTION and TASK in Verilog.	
Q3	(20 marks)	CO2, CO3
(a)	Explain the difference between: i. Blocking and Non-blocking assignement ii. begin-end and fork-join	
(b)	Design a clock with time period = 80ns and a duty cycle of 25% by using the always and initial statements. The value of clock at time = 0 should be initialized to 0.	
(c)	Write the Verilog description of T flip-flop and verify its working using suitable test bench.	
Q4	(20 marks)	CO4
(a)	What is UDP in Verilog? How many types of UDP are there? Explain the generic structure of UDP.	
(b)	Define a positive edge triggered D flip-flop with clear as a UDP. Signal clear is active low signal. Use shorthand notation wherever possible.	
(c)	Write the switch level Verilog description for AND gate.	
Q5	(20 marks)	CO5, CO6
(a)	Explain the basic structure of a testbench. Why testbench module don't have any port list?	
(b)	Write a test bench for the following: (i) 4×1 Multiplexer (ii) Full Adder	
(c)	Write a short note on Object Oriented Programming.	